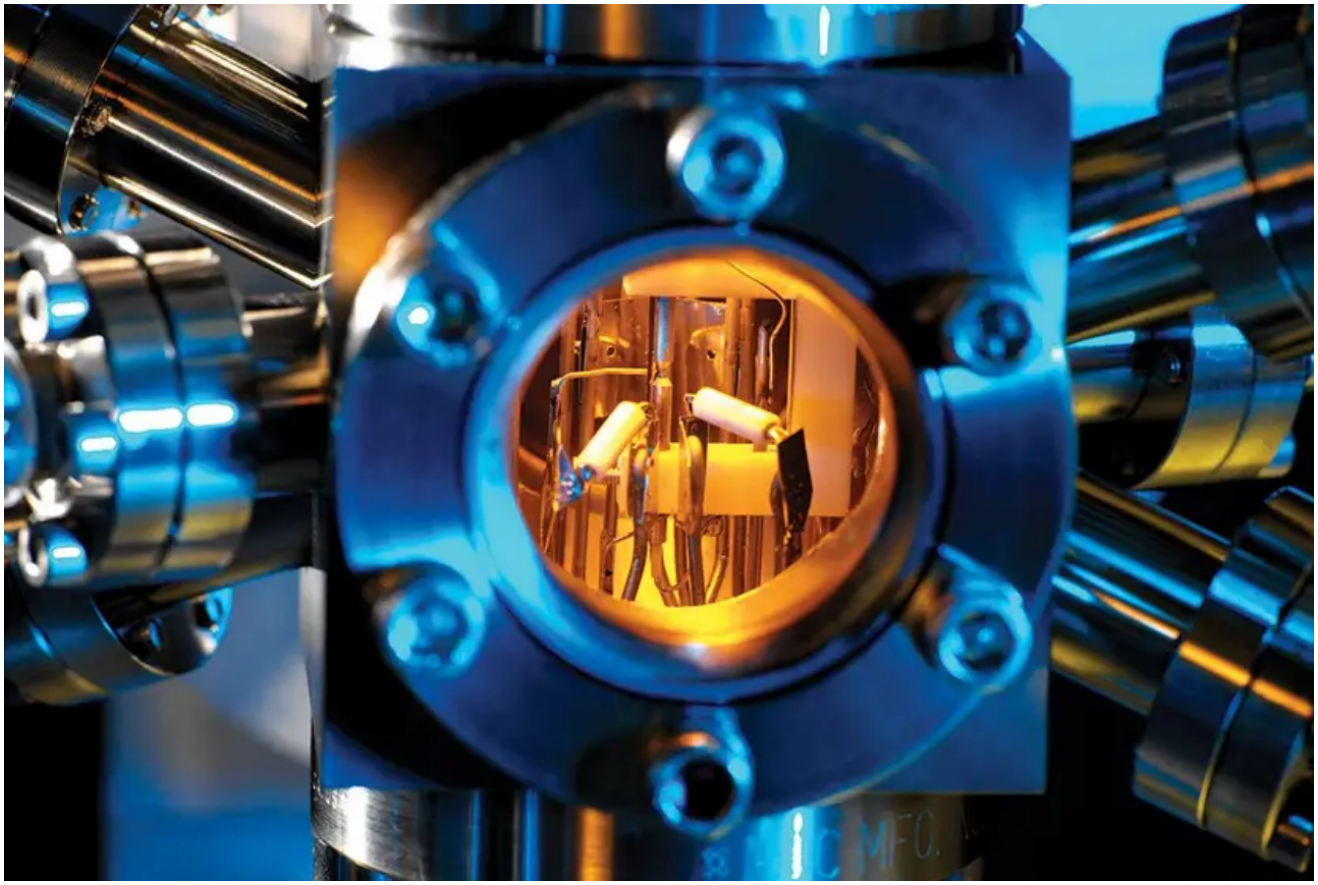

Atomic clocks.

Market volumes by application

<6 demo slides>

Semyon Fast, July 25

Aggregated TAM-SAM-SOM values for three product generations (table)

CSAC Generation	Total Addressable Market (TAM)	Serviceable Addressable Market (SAM)	Serviceable Obtainable Market (SOM), US
Gen 1 (MAC)	Industry Reviews: $250M - $300M Demand-Side Potential: $1.5B - $5.49B	$150M - $300M	$100M - $150M
Gen 2 (CSAC)	Industry Review: $23M - $400M Demand-Side Potential: $403.8M	$50M - $152.11M	$30M - $60.24M
Gen 3 (Ultra-Low SWaP)	Disruptive Target: $3B - $6.4B (Oscillator Market)	Projected: $230M - $1.2B (Estimate post-launch)	Projected: $10M - $600M (US Share, estimate post-launch)

TAM values are estimated using 2 methodlos: third-party industry reports (various), and demand-based calculations.
For demand, the required number of atomic clocks units was estimated for every major industry application.

Aggregated TAM-SAM-SOM values for three product generations (chart)

TAM values are estimated using 2 methodics: third-party industry reports (various), and demand-based calculations. For demand, the required number of atomic clocks units was estimated for every major industry application.

Annual units of atomic clock demand by industry application (table)

Application	Example Organizations	Annual Units	Product Gen
Wireless Base Stations	Huawei, Ericsson, Nokia	1.2M–1.5M	Gen 1
Wired Telecom (OMNBC)	Microchip, Weinberg	200K–300K	Gen 1
Dismounted Radios	L3Harris, Thales, GENS	100K	Gen 2
UAVs/UGVs	GA-ASI, Boeing, L3K, Textron	100K	Gen 2
Seismic (Oil & Gas)	Schlumberger, Halliburton	20K	Gen 2
Satellite Timing (LEO)	SpaceX, OneWeb, Wasar	20K	Gen 2/3
Power Grid PMUs	GE, Siemens, ABB	10K	Gen 1
Financial Services	Orolia, Weinberg, Microchip	3K	Gen 1/3
Robotics, Automotive, IoT	Bosch, NVIDIA, Mobileye	200K+	Gen 3
Total	-	2.200M	Gen 1, 2, 3

Source: Industry reports aggregation

Annual units of atomic clock demand by industry application (chart)

Source: industry reports aggregation

The atomic clock market is dominated by two key players who control **over 60% of the market share (by revenue)**

Company	Estimated Revenue	Market Share	Key Products
Orolia (Safran)	$120M	21%	Space clocks, RAFS
Microsemi (Microchip)	$250M	44%	CSAC, Rubidium
Frequency Electronics	$65.8M	12%	Space, Military applications
Teledyne (Instruments segment)	~$100 mln	18%	Cesium
Oscilloquartz (Adtran)	$26.8M	5%	Cesium, Grandmaster

Source: manual search for 2021 data, estimation of the atomic clock share in the revenue structure (for large cross-